

Multiple Choice

1. **Answer:** C

Options: A) IM - Trojans; B) Backdoor Trojans; C) SMS Trojan; D) Ransom Trojan

Note: An SMS Trojan is a type of malware that exploits vulnerabilities in mobile phones to send unauthorized text messages, often leading to unexpected charges on the user's phone bill.

2. **Answer:** A

Options: A) Escape queries; B) Interrupt requests; C) Merge tables; D) All of the above

Note: Escaping queries is essential in preventing SQL injection because it ensures that user input is treated as data rather than executable code, thus mitigating the risk of unauthorized SQL commands being executed.

3. **Answer:** A

Options: A) Receiver; B) Sender; C) Modulator; D) Translator

Note: The correct answer is "Receiver" because message confidentiality ensures that only the intended recipient can understand the transmitted information, thereby protecting it from unauthorized access or interception.

4. **Answer:** D

Options: A) Maltego; B) BurpSuite; C) Nessus; D) Wireshark

Note: Wireshark is a widely used network protocol analyzer that captures and displays packet data in real-time, making it an essential tool for wireless traffic sniffing in cybersecurity.

5. **Answer:** C

Options: A) Sender Site; B) Site; C) Receiver site; D) Conferencing

Note: The correct answer is "Receiver site" because encryption ensures that only the intended recipient, who possesses the decryption key, can access the original message, thereby maintaining confidentiality and data integrity during transmission.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: A hash function does not guarantee that a message has not been replaced with an identical message during transmission, as it is possible for two different messages to produce the same hash value, a phenomenon known as a hash collision.

7. **Answer:** False

Options: A) True; B) False

Note: Authentication is primarily concerned with verifying a user's identity through credentials such as passwords or biometric data, rather than restricting access based on user behavior and actions.

8. **Answer:** True

Options: A) True; B) False

Note: Wireless traffic sniffing involves capturing and analyzing data packets transmitted over wireless networks, which is essential for identifying security vulnerabilities, monitoring network performance, and conducting forensic investigations.

9. **Answer:** True

Options: A) True; B) False

Note: All of the listed authentication methods-secret questions, biometrics, and SMS codes-are recognized as valid techniques for verifying user identity in cybersecurity.

10. **Answer:** False

Options: A) True; B) False

Note: The Cotton Road was not primarily focused on counterfeit clothing but rather facilitated the illegal trade of a variety of goods, including drugs and hacking services, on the Dark Web.

Long Form Response

11. **Answer:** `def count_X(tup, x): | return count`

Note: The answer correctly identifies the need for a function that takes a tuple and an element as parameters and implies the use of a counting mechanism to determine the occurrences of the specified element within the tuple, which is essential for data analysis in cybersecurity contexts.

12. **Answer:** `def max_sum_subseq(A): | return A[0] | return look_up[n - 1]`

Note: correct because it identifies the need to consider non-adjacent elements in the array by implementing a dynamic programming approach, where the maximum sum is calculated based on previous maximum sums while ensuring that adjacent elements are not included in the sum.

End of Answer Key